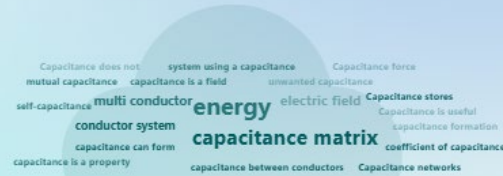


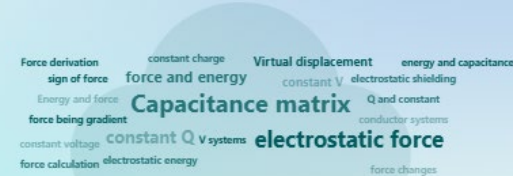
237 responses submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own...)



237 responses submitted

What is still unclear or confusing to you right now? (Responses should reflect your own understanding.)



Topic 2- Solutions of Electrostatic Problems

Outline

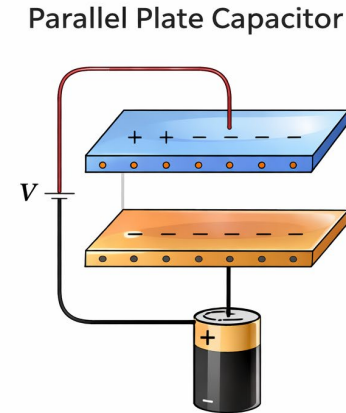
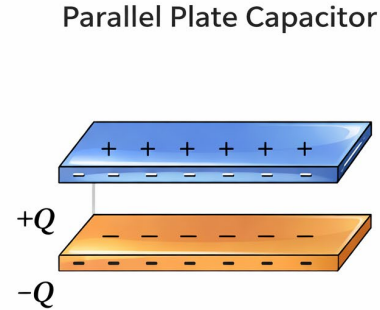
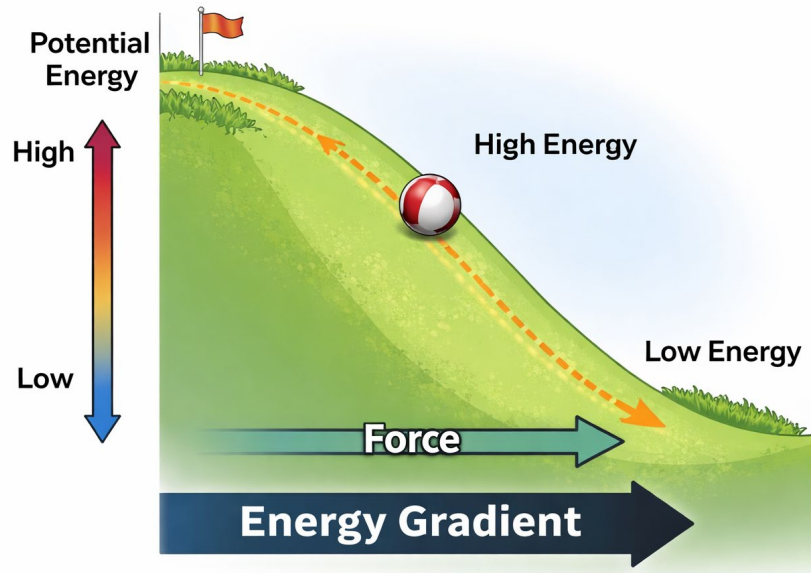
2.1 Electrostatic problems and their solution methods

2.2 Current in electrostatic problems

Chapters 4 and 5 of your textbook by D.K Cheng

Force is (negative) gradient of energy!

Force is simply the direction in which energy decreases fastest.

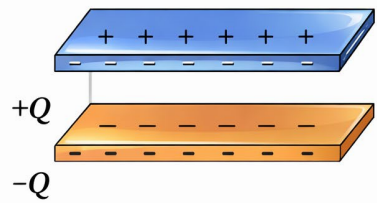


$$W_e = \frac{1}{2} QV \text{ J}$$

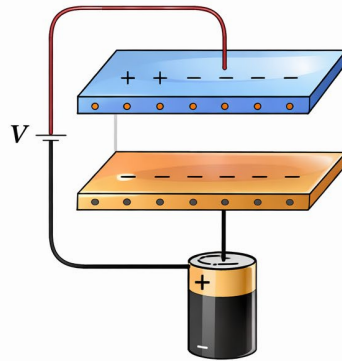
$$\vec{F}_e = \pm \vec{\nabla} W_e \rightarrow \vec{F}_x = \pm \left. \frac{dW_e}{dx} \right|_{Q \text{ or } V \text{ constant}}$$

- In a constant Q system, stored energy is spent on work done (e.g., to displace the force): negative gradient of W_e .
- In a constant V system (i.e., when there is a voltage source connected), the stored energy spent on work done is replenished by the source: the positive gradient of W_e .

Parallel Plate Capacitor



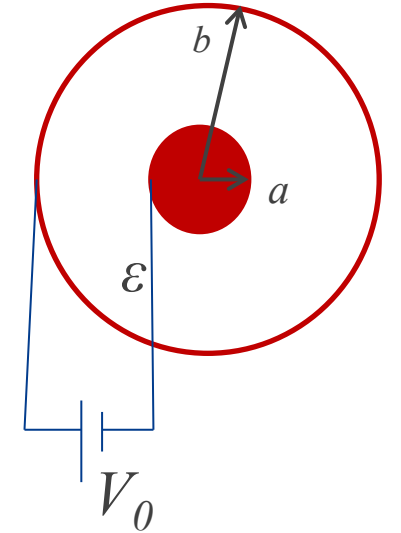
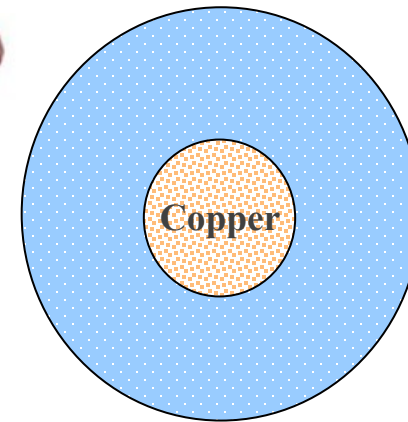
Parallel Plate Capacitor



How did we find $\mathbf{E}$ in these two cases?

$$\oint_S \vec{E} \cdot d\vec{s} = \frac{Q_{\text{enclosed}}}{\epsilon}$$

$$V_{ba} = -\int_a^b \vec{E} \cdot d\vec{l}$$

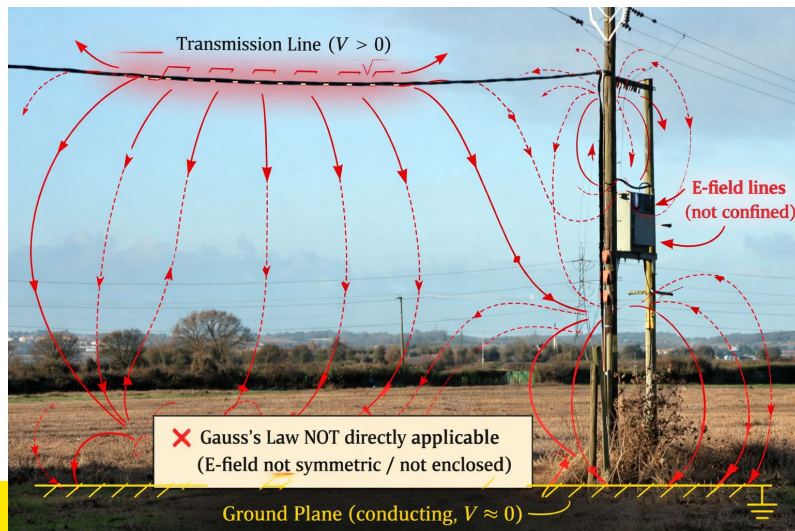


How did we find $\mathbf{E}$ in this case?

$$\vec{E} = \left| \frac{\rho_l}{2\pi\epsilon r} \right| \vec{a}_r \quad \text{V/m}$$

If we are given the potential, instead of the charge density, **how will we find $\mathbf{E}$?**

How to find the $\mathbf{E}$ when the effect of the ground cannot be ignored, i.e., when symmetry is broken, and Gauss's law cannot be applied directly?



When Gauss's Law is not Enough

Charges known $\rightarrow$ Use Gauss's law $\rightarrow$ Find $\mathbf{E}$ and then V

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon} \quad \vec{E} = -\vec{\nabla} V$$

Voltage V known $\rightarrow$ how to find $\mathbf{E}$

$$\vec{\nabla}^2 V = -\frac{\rho}{\epsilon} \rightarrow \text{Find } \vec{E}$$

We are about to learn how Engineers solve this.

Learning goals

After today's lecture you will be able to



Calculate electric field $\mathbf{E}$ from potential V using Laplace's and Poisson's equations.



Analyse and calculate the electric field $\mathbf{E}$ and capacitance C using the Method of Image when Gauss's law is inapplicable.

Poisson's and Laplace's Equations

We know : $\vec{\nabla} \bullet \vec{E} = \frac{\rho}{\epsilon} \quad (1), \quad \vec{E} = -\vec{\nabla} V \quad (2)$

Substitute (2) in (1), $\vec{\nabla} \bullet (-\vec{\nabla} V) = \frac{\rho}{\epsilon} \rightarrow \vec{\nabla} \bullet \vec{\nabla} V = -\frac{\rho}{\epsilon} \quad (3)$

$\vec{\nabla} \bullet \vec{\nabla} = \vec{\nabla}^2$ is called the Laplacian operator.

(3) is written as, $\vec{\nabla}^2 V = -\frac{\rho}{\epsilon}$ **Poisson's equation.**

If ρ is zero, i.e. a charge free area, $\vec{\nabla}^2 V = 0$ **Laplace's equation.**

80% of engineering electrostatic problems use Laplace's equation.

$$\vec{\nabla}^2 V = -\frac{\rho}{\epsilon}$$

Cartesian co-ordinate system,

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$$

Cylindrical co-ordinate system,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$$

Poisson's and Laplace's equations are **scalar** but second-order partial differential equations and always have unique solutions.

Solution methods

How to solve Poisson's and Laplace's equations?

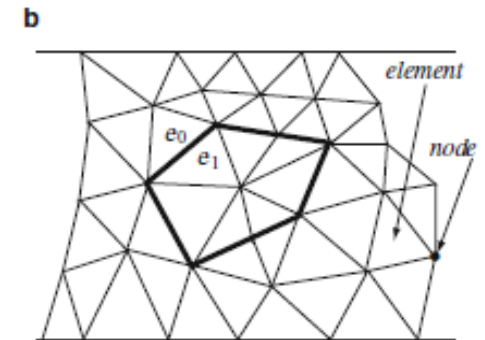
Using any method of solving second-order partial differential equations – analytical or numerical.

- **Analytical methods**

1. *Direct integration* (for one-dimensional problems)
2. *Method of Image* (for solving Poisson's equation of simple geometry)
3. *Separation of variables* (for solving Laplace's equation, 2-D geometry).

- **Numerical methods (3/2-D complex geometry)**

Finite Element Method (FEM e.g., software ANSYS Maxw

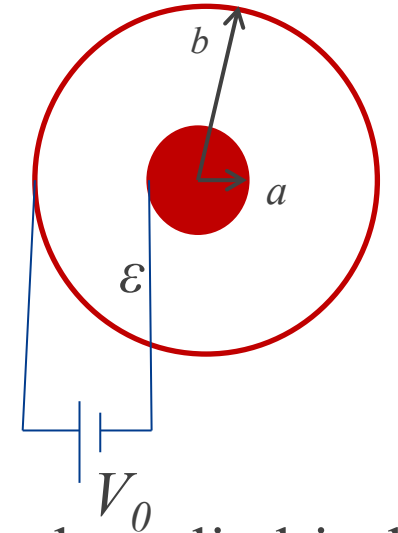


How Do I Know Which Method To Use?

Given	Use
High symmetry + known charge (or density)	Gauss's Law
Charge-free region + boundary voltages known	Laplace Equation • Direct integration, (1-D) • Method of separation (2-D)
Conducting plane or simple geometry	Method of Image
Complex geometry (2-D or 3-D)	Laplace/Poisson's Equations • Numerical methods (FEM)

Example 1: Direct integration

Two co-axial conductors as shown in the figure has a potential difference between them V_0 . The conductors are assumed to be very long. Find the $\mathbf{E}$ between the conductors.



Step 1: What do we know?

- Long cylindrical conductor
- No volume charge density ρ in the study region $\rightarrow$ Laplace's equation in the cylindrical coordinate.

$$\vec{\nabla}^2 V = 0 \quad \rightarrow \quad \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

But V varies only radially $\rightarrow$ 1-D problem. $\frac{1}{r} \frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$

$$\text{Simplify, } \frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$$

Step 2: Direct integration

$$\frac{d}{dr} \left(r \frac{dV}{dr} \right) = 0$$

Integrating once, $r \frac{dV}{dr} = k_1 \rightarrow \frac{dV}{dr} = \frac{k_1}{r}$

Integrating again, $V(r) = k_1 \ln(r) + k_2$ where, k_1, k_2 : integration constants

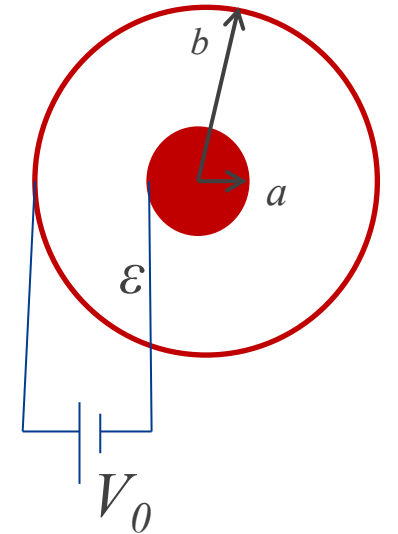
k_1 and k_2 are found from two boundary conditions of potential.

Step 3: Apply boundary conditions

$$V(r = a) = -V_0 \text{ and } V(r = b) = 0 \quad (\text{Check how the battery is connected.})$$

$$V(a) = k_1 \ln(a) + k_2 = -V_0,$$

$$V(b) = k_1 \ln(b) + k_2 = 0$$



Step 4: Solve for k_1 and k_2

$$k_1 \ln(b) + k_2 = 0 \rightarrow k_2 = -k_1 \ln(b)$$

$$k_1 \ln(a) + k_2 = -V_0 \rightarrow k_1 \ln(a) - k_1 \ln(b) = -V_0 \rightarrow k_1 = \frac{V_0}{\ln\left(\frac{b}{a}\right)} \quad \text{and} \quad k_2 = -k_1 \ln(b) = \frac{-V_0}{\ln\left(\frac{b}{a}\right)} \ln(b)$$

Step 5: Find $V(r)$ and $\mathbf{E}(r)$

$$V(r) = k_1 \ln(r) + k_2 \quad \rightarrow \quad V(r) = \frac{V_0}{\ln\left(\frac{b}{a}\right)} [\ln(r) - \ln(b)], \quad a \leq r \leq b$$

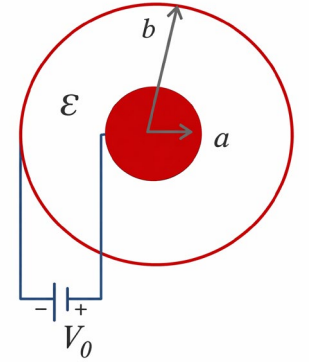
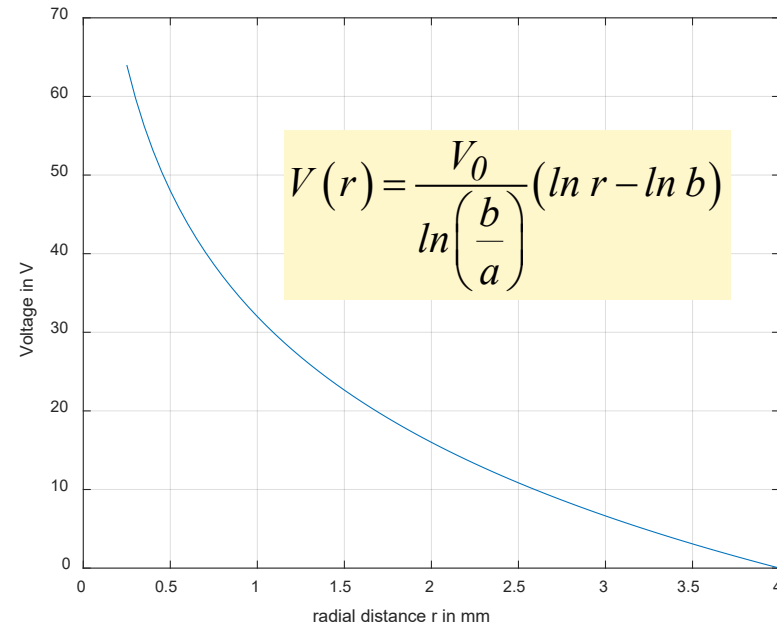
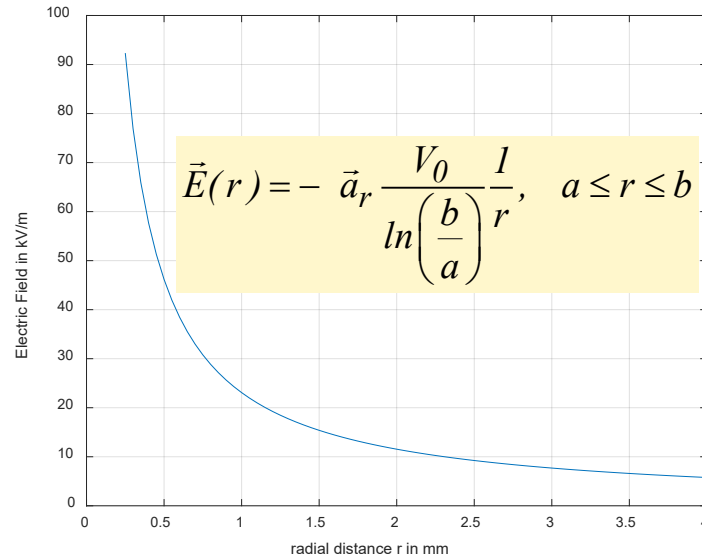
Electric field is $\vec{E}_r = -\vec{\nabla} V(r)$

$$\vec{E}_r = -\vec{a}_r \frac{dV(r)}{dr} = -\vec{a}_r \frac{d}{dr} \left(\frac{V_0}{\ln\left(\frac{b}{a}\right)} (\ln(r) - \ln(b)) \right) = -\vec{a}_r \frac{V_0}{\ln\left(\frac{b}{a}\right)} \frac{1}{r}$$

Engineering interpretation:

Let's take an example cable, $a = 0.25$ mm, $b = 4$ mm, $V_0 = 64$ V.

Plot of $\mathbf{E}(r)$ and $V(r)$:



$\mathbf{E}$ is strongest at the minimum radius, V varies logarithmically with r .

- (i) Observing the variation of E_r , we can recommend a dielectric material for the cable based on a safety factor and dielectric strength (for example: Polyethylene (PE): 450kV/m, Plastic: 400kV/m, Polytetrafluoroethylene (PTFE): 1.7MV/m, Air: 3 MV/m)
- (ii) Observing the variation of V , we can predict the potential at any radial distance from the conductor.
- (iii) For a given voltage and dielectric, we can also determine safe thickness

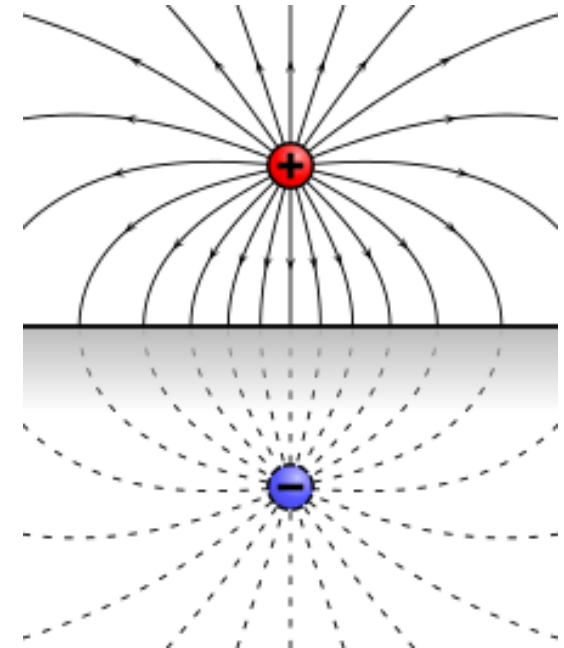
Method of images



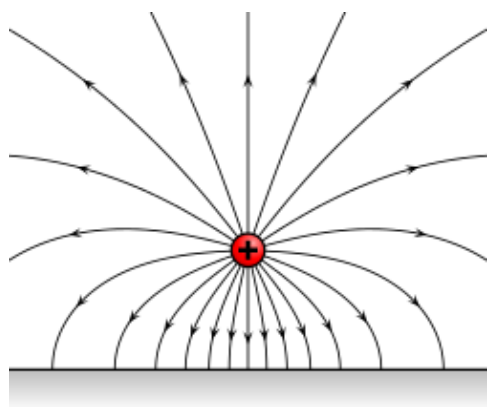
When you stand in front of a mirror:

- You see your reflection.
- The reflection is not real.
- But it looks exactly positioned to satisfy symmetry.

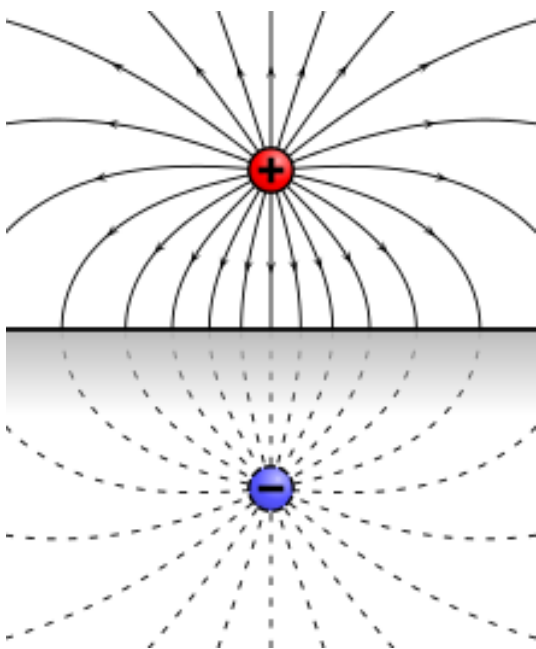
What if electric fields behaved the same way near conductors?



A charge near a conducting plane



Ground plane



We want to find $\mathbf{E}$ and V in the space around.

It requires solving 3-D Poisson's equations with complicated boundary conditions.

The alternative is the Method of Images.

Ground plane becomes $V=0$ *conducting surface* automatically and behaves like a mirror.

Why Can We Do This?

There is a powerful principle behind the solutions of Partial Diff. Eqn (Uniqueness theorem)

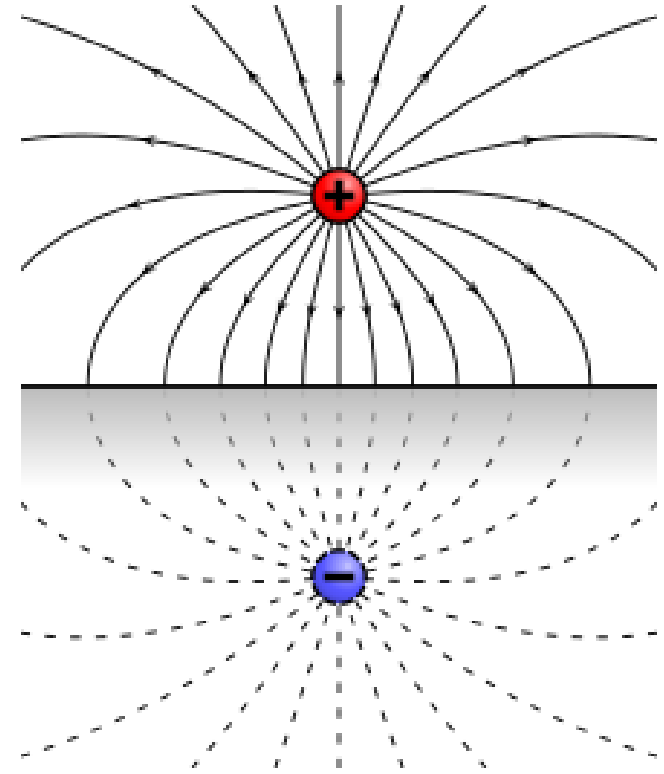
If a solution satisfies Poisson/Laplace equation and boundary conditions, then it is the unique correct solution.

Our image charge arrangement satisfies these.

The Big Idea

The Method of Images replaces a complicated boundary problem with a simpler charge superposition problem.

Conductors act like mirrors for electric fields.



Important Notes

Image charges are mathematical tools.

They do not exist physically.

They only reproduce the correct field in the region of interest.

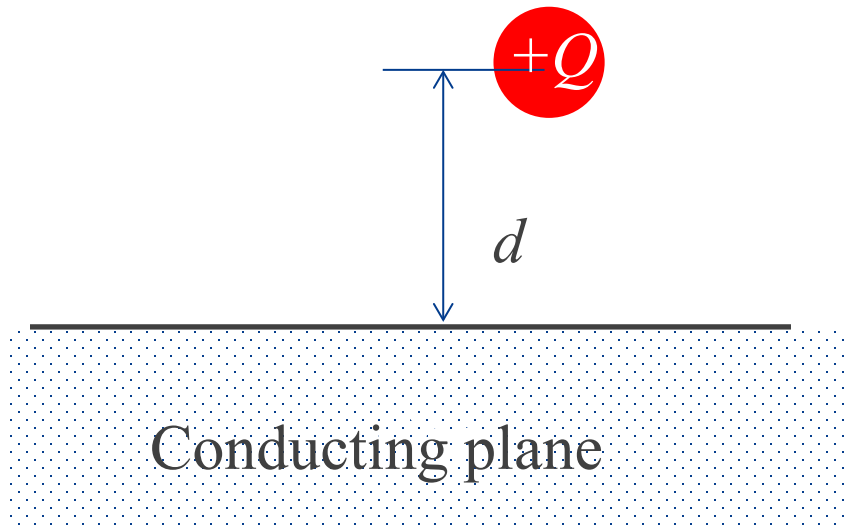
We never care about the region below the conductor (in the example below, the ground plane). We only care about the physical region.

Example 2

Find the potential V at point $P(x,y,z)$ for a point charge Q , which is d meters above a large conducting plane. The conducting plane is at zero potential.

Let us list the boundary conditions that our Method of Images solution must satisfy:

$P(x,y,z)$



$$V(x, 0, z) = 0; \text{ (on the plane)}$$

$$V \rightarrow \frac{Q}{4\pi\epsilon_0 R} \text{ (close to the charge)}$$

$$V = 0 \text{ (any point far from the charge)}$$

$$V(x, y, z) = V(-x, y, z);$$

$$V(x, y, z) = V(x, y, -z)$$

Solution

Apply the method of images and define radial distances R_+ and R_- from the charge Q and its image $-Q$ to point P .

Potential V at point $P(x,y,z)$ is due to the two point charges at $Q(0,d,0)$ and $-Q(0,-d,0)$:

$$V_p(x, y, z) = V_+ + V_- = \frac{Q}{4\pi\epsilon_o} \left[\frac{1}{R_+} - \frac{1}{R_-} \right]$$

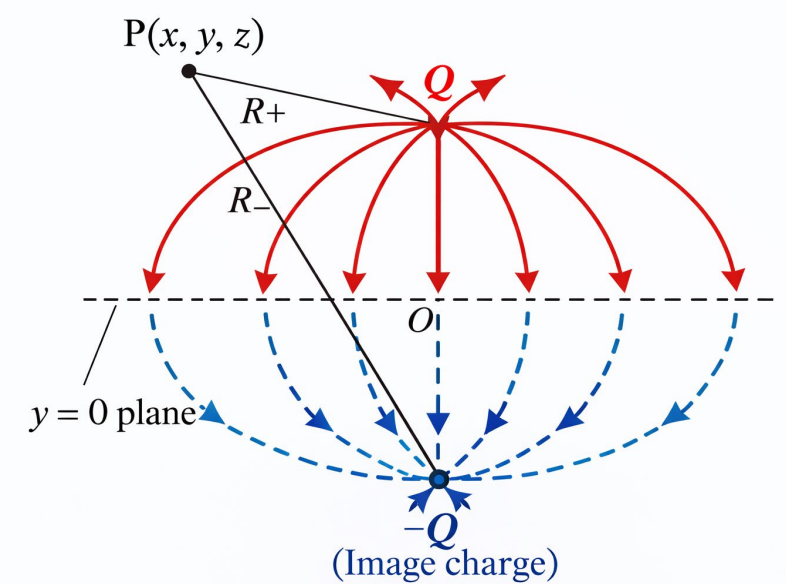
where,

$$R_+ = \sqrt{x^2 + (y-d)^2 + z^2} \quad R_- = \sqrt{x^2 + (y+d)^2 + z^2}$$

Let see whether the solution satisfies all boundary conditions and Poisson's equation:

$$\vec{\nabla}^2 V = -\frac{Q}{\epsilon_o}$$

Note that $\mathbf{E} = -\vec{\nabla}V_P$.



Boundary conditions:

$V(x, 0, z) = 0$; (on the plane)

$V \rightarrow \frac{Q}{4\pi\epsilon_o R}$ (close to the charge, $R_- \rightarrow \infty$)

$V = 0$ (any point far from the charge)

$V(x, y, z) = V(-x, y, z)$;

$V(x, y, z) = V(x, y, -z)$

Pause & Reflect

(Before we take a short break...)

✎ **Take 1 minute to jot down:**

One idea you just learned

One step you could explain to a friend

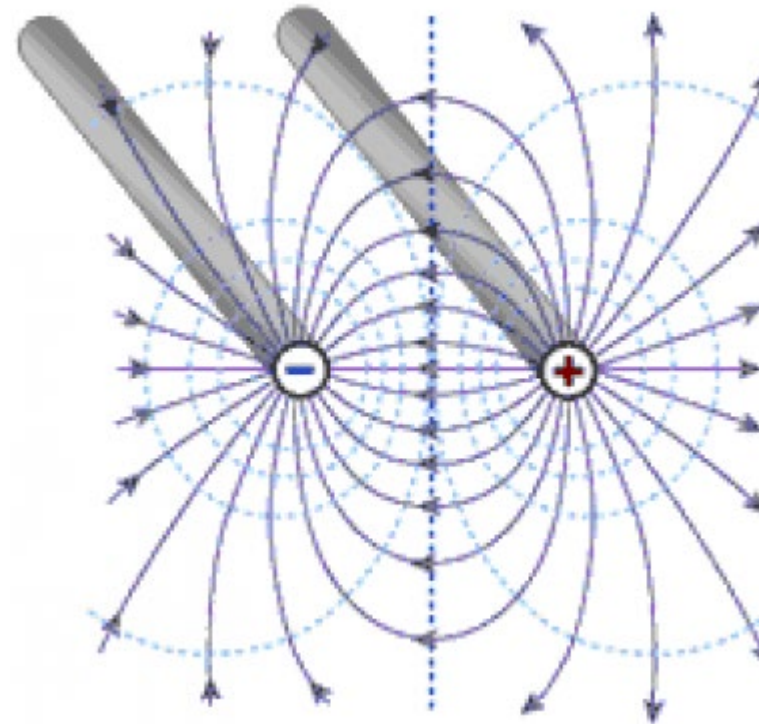
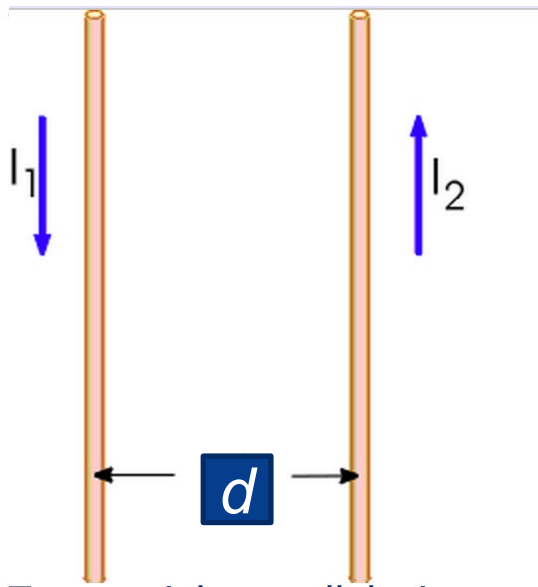
One word that describes how confident you feel right now
(scale: 😞 *unsure* → 😊 *okay* → 😄 *confident*)

💡 *This is for you — no submission.*



Applications of the Method of Images to cylindrical conductors

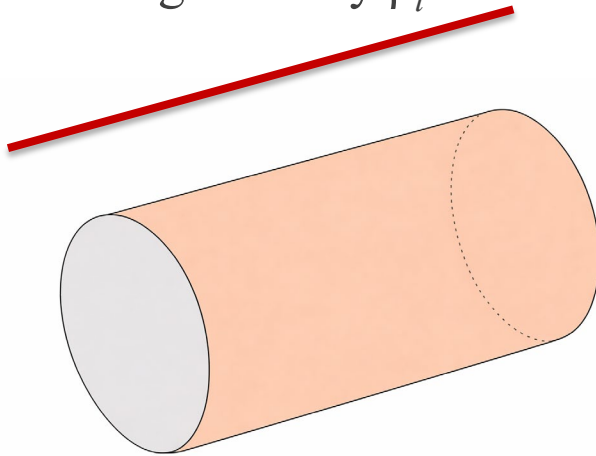
Recall, two parallel conductors system where we assume distance between them is very large and electric field of one does not alter the field of the other. Using method of image, we do not need to assume $d \gg a$. We can find field and capacitance between two conductors that are close to each others.



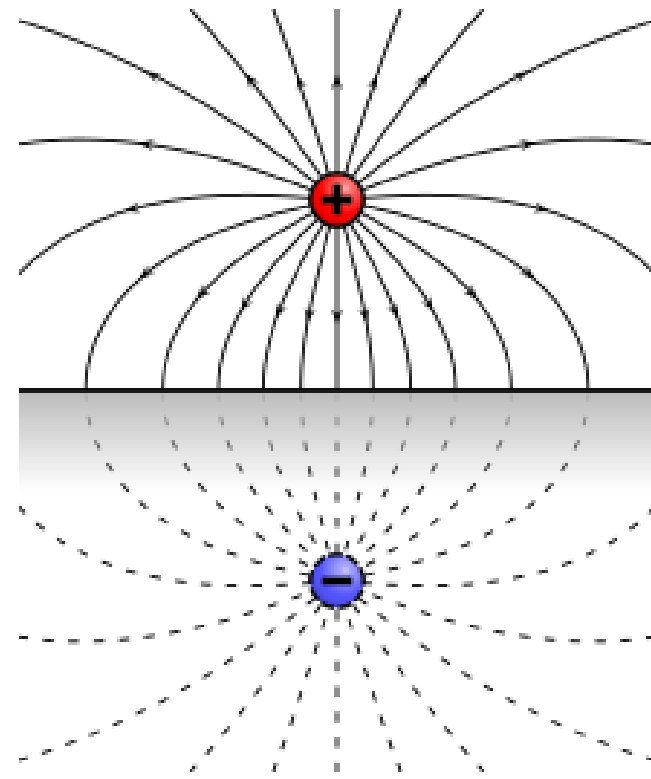


Conducting plane

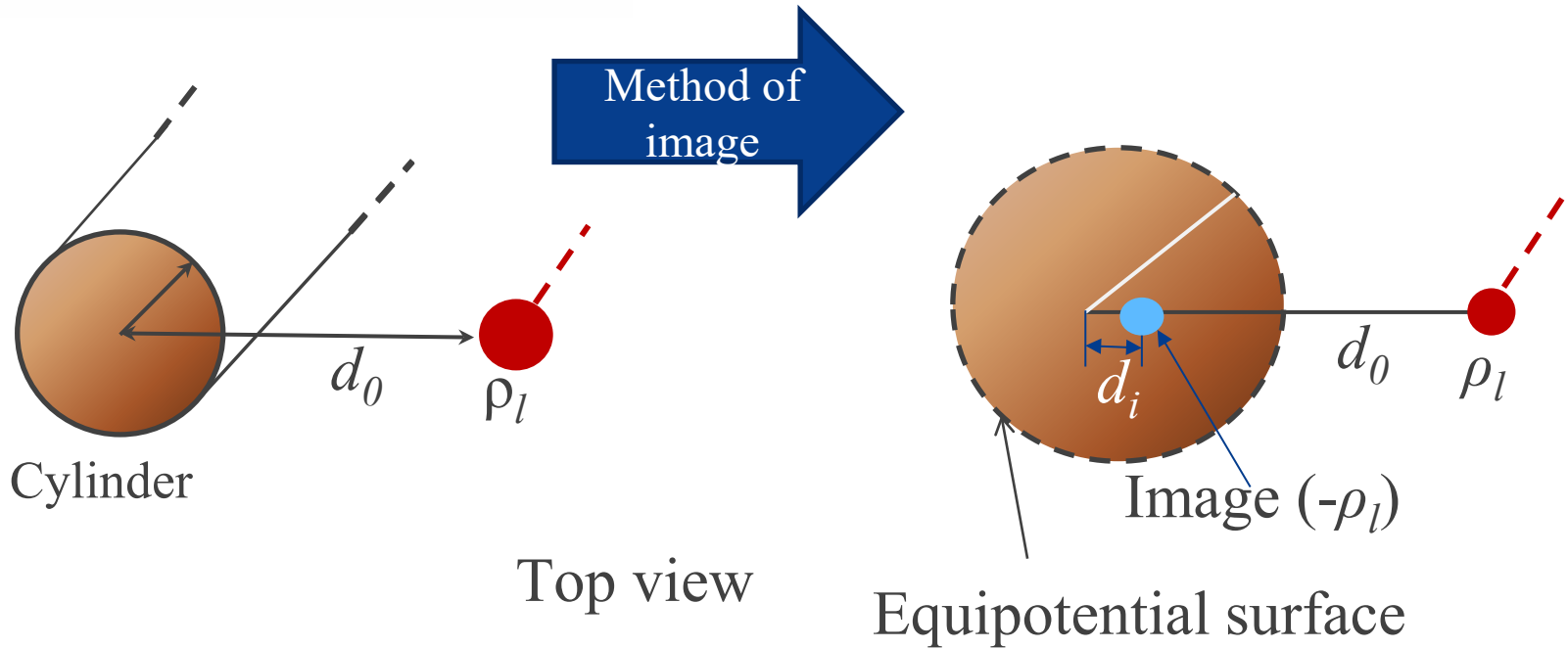
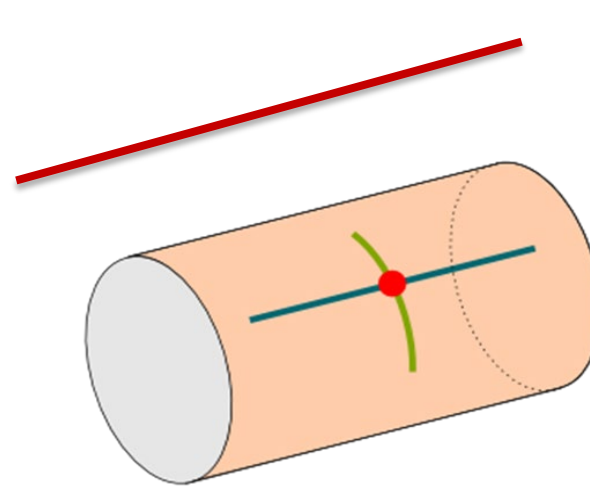
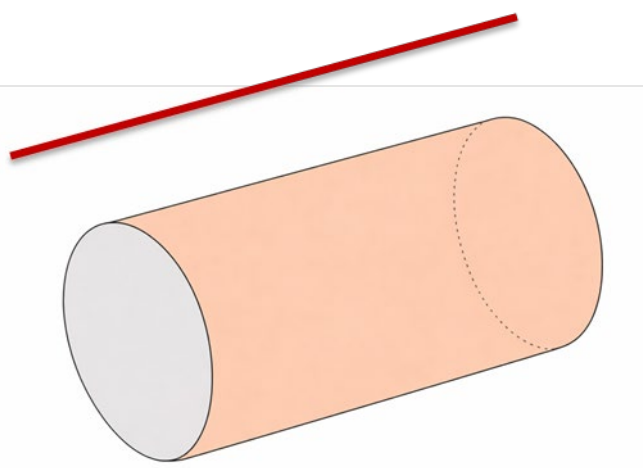
Line charge density ρ_l



Method of image

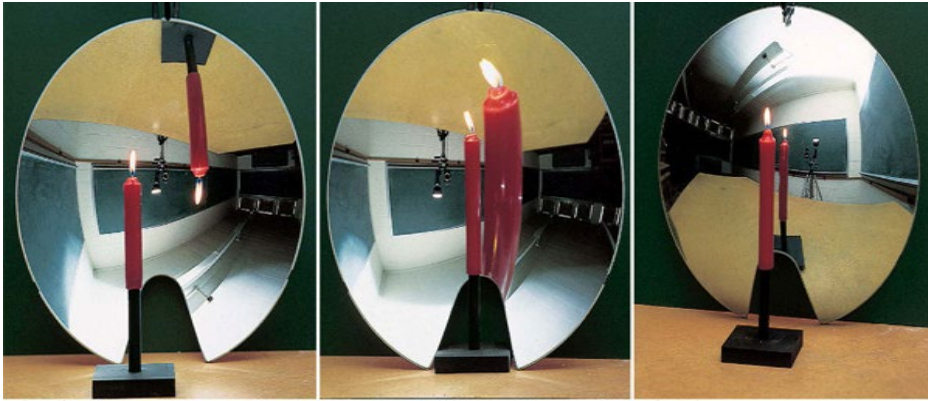


Can we apply method of image to a conductor with a curved geometry such as a cylindrical conductor ?



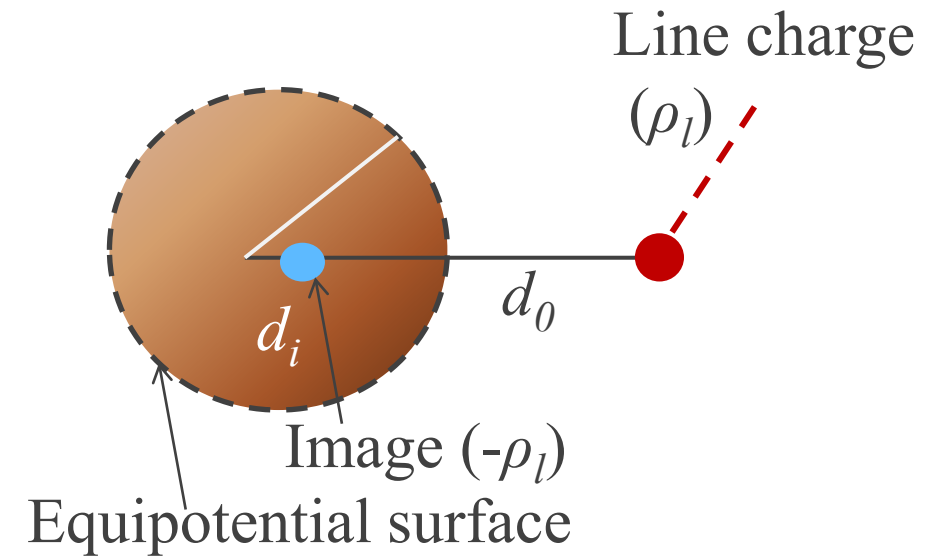
Step A - What must be true

- For conductor: surface must be equipotential (i.e., potential is the same at every point on the surface)
- Image of ρ_l must be a negative parallel line charge inside the cylinder at a distance d_i from the centre of the cylinder when d_0 is the distance to the line charge from the same centre.



Flat, convex and concave mirrors

Unlike a flat mirror, in a curved mirror distance between the mirror and image is not the same as the source and the mirror, i.e., $d_i \neq d_0$



Step B – Strategy

- What should be d_i for a line charge image?
- For the method of images, the cylindrical surface of the conductor must be equipotential. How to prove that the surface is equipotential?

Proof: the cylindrical surface is equipotential

Electric field intensity at a point M on the surface of the conductor which is at distance r from $+\rho_l$

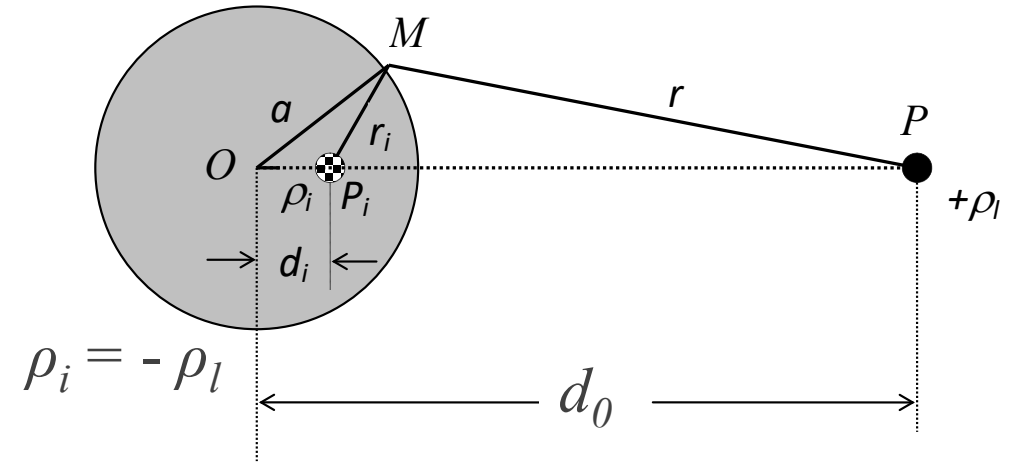
$$\vec{E}_r = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r$$

Potential V due to $+\rho_l$ at M w.r.t a reference at r_0 (not shown):

$$V = -\int_{r_0}^r \vec{E}_r \bullet d\vec{r} = -\frac{\rho_l}{2\pi\epsilon_0} \int_{r_0}^r \frac{1}{r} dr = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

The potential at M due to the image charge $-\rho_l$: $V_i = \frac{-\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r_i}$

Total potential at M by superposition, $\therefore V_M = V + V_i = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r} - \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_0}{r_i} = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_i}{r}$



$V_M = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{r_i}{r}$ For equipotential, point M anywhere on the surface must give the same value.

What will make V_M constant anywhere on the surface?

(If r_i/r is a constant, the cylindrical surface is at an equipotential.)

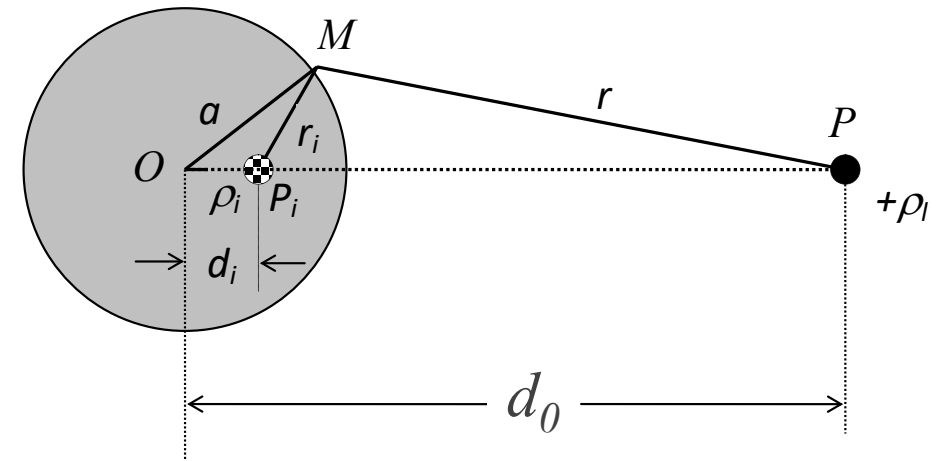
Note that the triangle OMP_i and OMP share one angle ($\angle MOP = \angle MOP_i$) and one side (OM).
If they are similar triangles,

$$\frac{P_iM}{PM} = \frac{OP_i}{OM} = \frac{OM}{OP} \rightarrow \frac{r_i}{r} = \frac{d_i}{a} = \frac{a}{d_0}$$

Since, $r_i/r = a/d_0$, the potential V_M is

$$V_M = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{d_0}$$

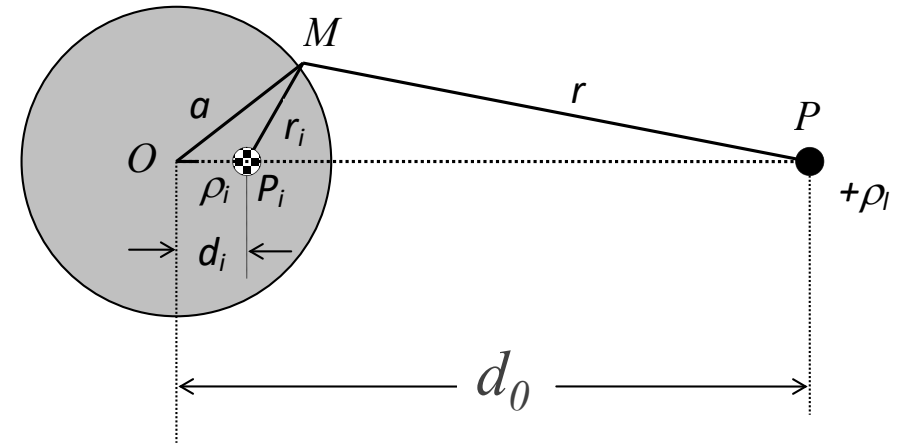
$$\text{Also } d_i = \frac{a^2}{d_0}$$



$$\frac{r_i}{r} = \frac{d_i}{a} = \frac{a}{d_0} \quad (\text{Similar triangles})$$

$$d_i = \frac{a^2}{d_0}$$

$$V_M = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{d_0}$$



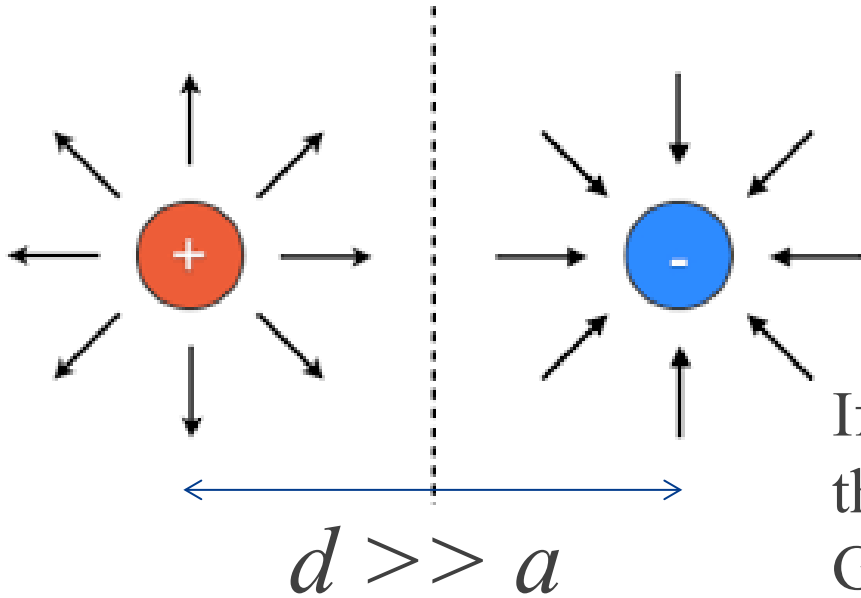
Equipotential surface,
Method of image applicable !

Thus, if the image line charge is located at $d_i = a^2/d$, r_i/r remains constant as a/d_0 , i.e., the cylindrical surface is equipotential (any point on the surface has the same constant potential.) This proves that an image line charge and equipotential cylindrical surface can replace a cylindrical conducting body using the image method. The condition of $d_i = a^2/d$ is satisfied only when the two triangles formed are similar triangles.

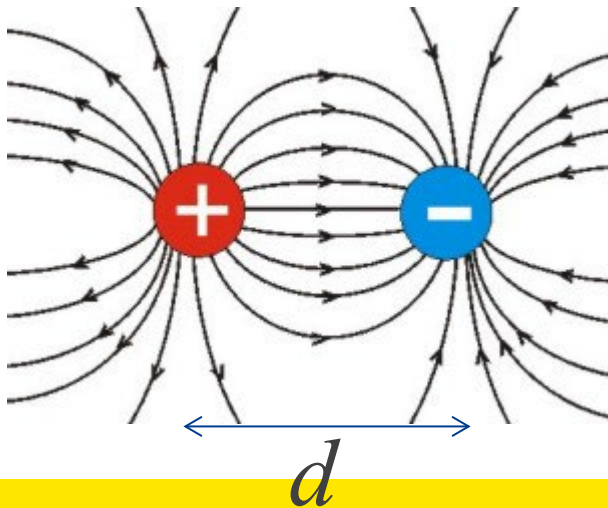
Why the method of images of a cylindrical conductor matters:

To use Gauss's law to find C between lines, we assumed $d \gg a$

$$C \approx \frac{\pi\epsilon_0}{\ln \frac{d-a}{a}} \text{ F/m}$$



If the conductors are not very far apart, we need to consider the proximity effect on the field distribution, and hence, Gauss's law cannot be applied; instead, we must use the method of images of the cylindrical conductor.

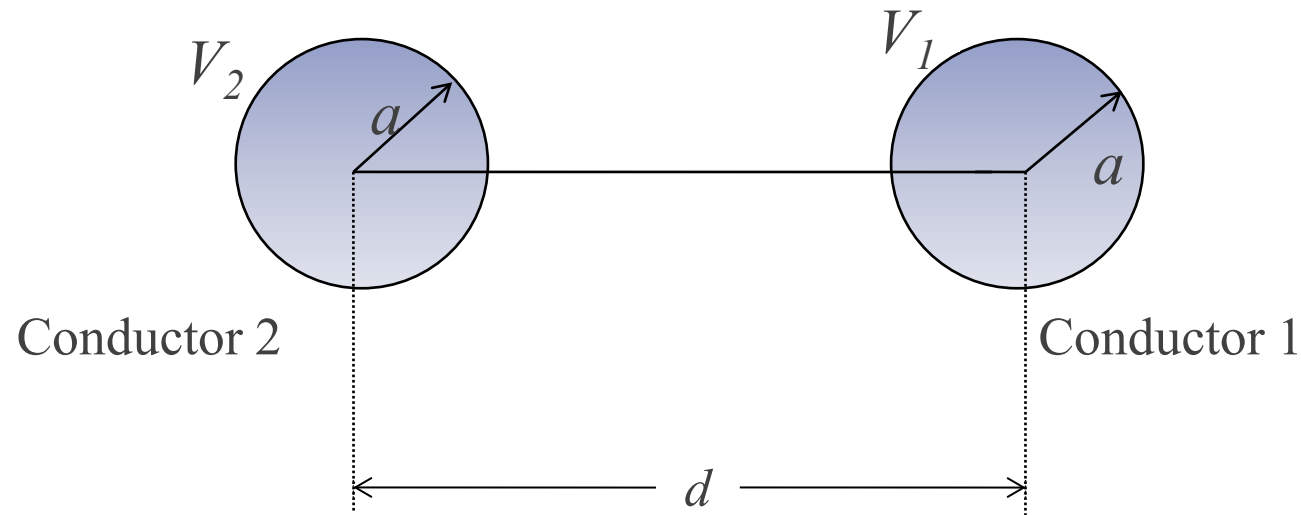


Now we can eliminate “ $d \gg a$ ” assumption.

We can now model real-world transmission lines and pcb tracks.

Example 3

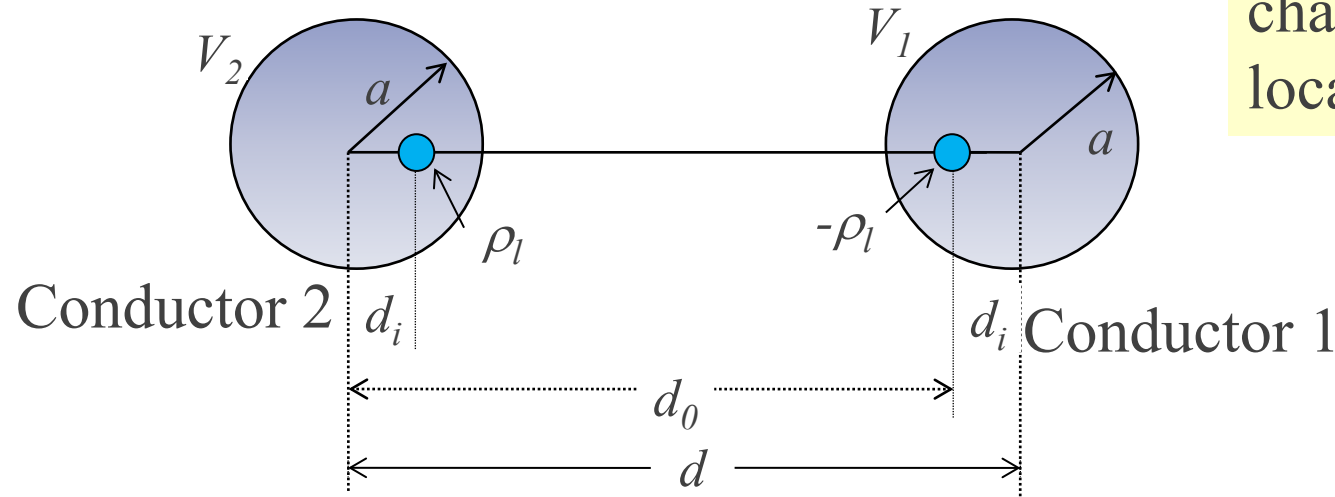
Find Capacitance per unit length between two cylindrical conductors at close proximity (i.e. d and radius a are of comparable value).



If we can compute $V = V_1 - V_2$ using the method of images $\rightarrow$ we can compute C

Solution

We are no longer assuming uniform charge density all around. Hence location of ρ_l should be defined.



d : is distance between centre of conductor 2 to charge $-\rho_l$ in conductor 1.

By method of image, $-\rho_l$ is image of $+\rho_l$ and vice a versa.

Hence, the potential at the surface of conductor 1 due to ρ_l located at distance d_i from the centre of the conductor 2 is

$$V_1 = \frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{d_0}$$

Similarly, potential at the surface of conductor 2 due to $-\rho_l$

$$V_2 = -\frac{\rho_l}{2\pi\epsilon_0} \ln \frac{a}{d_0}$$

$$\therefore V_{21} = V_2 - V_1 = \frac{\rho_l}{\pi\epsilon_0} \ln \frac{d_0}{a}$$

Hence, $C = \frac{\rho_l}{V_{21}} = \frac{\pi\epsilon_0}{\ln \frac{d_0}{a}} \text{ F/m}$

d_0 can be found as below:

$$d_0 = d - d_i$$

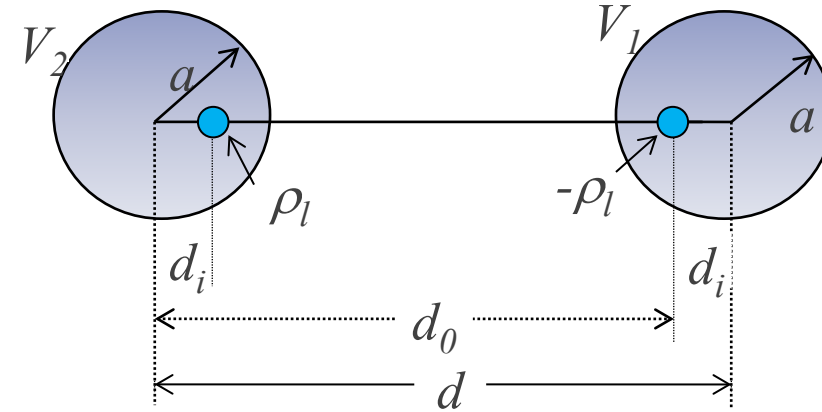
But, $d_i = \frac{a^2}{d_0}$

$$\therefore d_0 = d - \frac{a^2}{d_0} \rightarrow d_0^2 - dd_0 + a^2 = 0$$

Solving this quadratic equation, $d_0 = \frac{d \pm \sqrt{d^2 - 4a^2}}{2}$

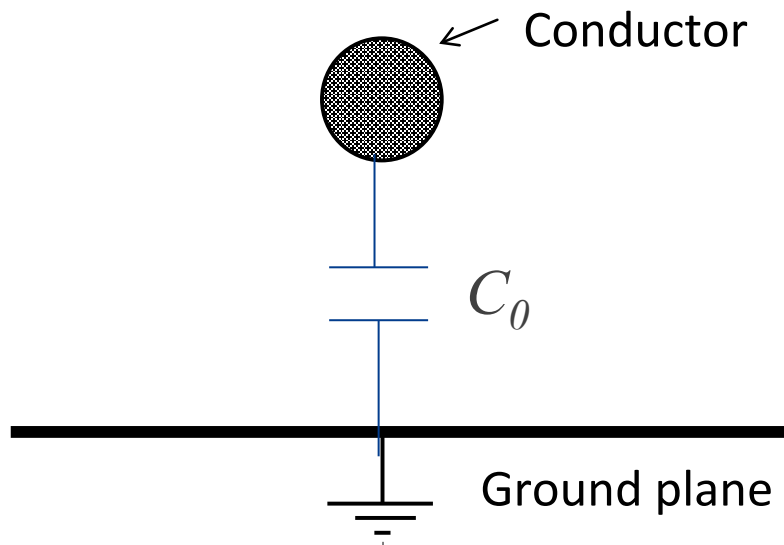
Sub d_0 in $C = \frac{\pi\epsilon_0}{\ln \frac{d_0}{a}}$

$$\therefore C = \frac{\pi\epsilon_0}{\ln \left[\left(\frac{d}{2a} \right) + \sqrt{\left(\frac{d}{2a} \right)^2 - 1} \right]} \quad \text{F/m}$$



*Use this formula, to find capacitance of closely located parallel conductors i.e. d is not too large compared to a .

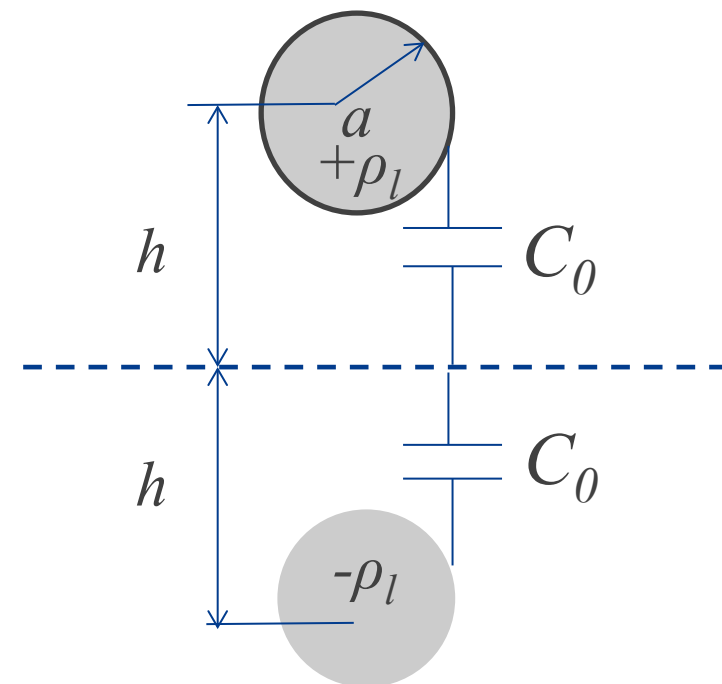
Practice Example



$$d=2h$$

$$C = \frac{\pi\epsilon_0}{\ln\left[\left(\frac{d}{2a}\right) + \sqrt{\left(\frac{d}{2a}\right)^2 - 1}\right]} \text{ F/m}$$

$$1/C = 1/C_0 + 1/C_0 \rightarrow C_0 = 2C$$



With the help of the image charge, find the capacitance per meter between a conductor of radius $a = 10 \text{ mm}$ which runs parallel to a ground at height $h = 10\text{m}$.

$$\text{Ans: } C_0 = 7.32\text{pF/m}$$

Limitation of Method of image

It cannot be used,
when

the geometry of the system is too complex to find images,

the conducting body and the ground plane are not of infinite extent,

the potential V at the boundary is the only known/measurable quantity.

Before You Go... (1 minute reflection)

This reflection will count toward optional bonus marks.

0 response submitted

What is the most important idea you are taking away from today's lecture? (Responses should reflect your own understanding.)

Scan the QR or use link to join



<https://forms.office.com/r/vJs4eSWdQH>

Copy link

Wordcloud All responses

1 of 3

Review and further practice (Recommended)

What You Must Be Able To Do For Exams

Students should be able to:

1. Identify the correct method.
2. Reduce the Laplace equation using symmetry (direct integration).
3. Apply boundary conditions.
4. Compute E from V .
5. Explain why the image method works conceptually.
6. Compute C between conductors spaced closely by using the method of images
7. Compute C between conductor and ground plane using the method of images

Check the workouts of Examples 4-1 and 4-2 of the textbook of D.K. Cheng.

You may practice P4-10, P4-11, P4-13